

Year 7 • Chapter 9 • The number plane: answers (teacher copy)

9.01 Grid references on maps (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: A1 → Pool
- Column 1, E2: A4 → Farm
- Column 2, E1: Bus to Fire → Fire is at B6; 3 left, 4 up
- Column 2, E2: Oval to Shop → Shop is at B2; 5 left, 2 down
- Column 3, E1: Start at the Bank. Go 5 right and 5 up.
Where are you? → start: C1; across +5 → H; up/down +5 → 6; H6: the Hotel
- Column 3, E2: Start at the Shop. Go 5 right and 1 up.
Where are you? → start: B2; across +5 → G; up/down +1 → 3; G3: the Cafe

Questions (in order: Easy, Medium, Challenging)

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|----------|-------------------------|-------------------------|
| 1. Post | 13. Cafe | 23. H2; 4 right, 4 down |
| 2. Zoo | 14. Oval | |
| 3. Beach | 15. Shop | 24. G3; 4 right, 2 up |
| 4. Hotel | 16. Club | 25. F6; 2 right, 3 up |
| 5. Gym | 17. C4; 3 left, 3 up | 26. A1; 4 left, 4 down |
| 6. Vet | 18. D6; 3 right, 2 up | |
| 7. Park | 19. C1; 1 right, 5 down | 27. F1, the Park |
| 8. Mall | 20. D3; 4 left, 3 down | 28. C1, the Bank |
| 9. Wharf | 21. E5; 3 right, 3 up | 29. C4, the Mall |
| 10. Bus | 22. A6; 2 left, 2 up | 30. E2, the Bus |
| 11. Fire | | 31. D3, the Gym |
| 12. Bank | | 32. A1, the Pool |

9.02 Points in the first quadrant (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: point A → (1, 4)
- Column 1, E2: point B → (4, 0)
- Column 2, E1: (9, 6) and (9, 0) → same x, so count up: 6 – 0; = 6 units
- Column 2, E2: (8, 2) and (8, 8) → same x, so count up: 8 – 2; = 6 units
- Column 3, E1: (2, 4), (9, 4), (9, 9) → same x as (2, 4): x = 2; same y as (9, 9): y = 9; fourth corner (2, 9)
- Column 3, E2: (4, 1), (7, 1), (7, 9) → same x as (4, 1): x = 4; same y as (7, 9): y = 9; fourth corner (4, 9)

Questions (in order: Easy, Medium, Challenging)

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|-----------|-------|------------|
| 1. (0, 3) | 10. 7 | 19. (0, 9) |
| 2. (8, 2) | 11. 7 | 20. (1, 5) |
| 3. (2, 4) | 12. 5 | 21. (1, 7) |
| 4. (0, 4) | 13. 7 | 22. (4, 7) |
| 5. (8, 8) | 14. 2 | 23. (2, 7) |
| 6. (0, 7) | 15. 5 | 24. (3, 6) |
| 7. (6, 7) | 16. 7 | 25. (4, 5) |
| 8. (0, 2) | 17. 2 | |
| 9. 3 | 18. 2 | |

9.03 The number plane (booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: (–7, 4) → 2nd
- Column 1, E2: (4, –3) → 4th
- Column 2, E1: point A → (–4, –3); 3rd quadrant
- Column 2, E2: point B → (–4, 5); 2nd quadrant
- Column 3, E1: Reflect (4, 5) in the y-axis. → y-axis: the x-coordinate changes sign; (4, 5) → (–4, 5); quadrant: 1st → 2nd
- Column 3, E2: Reflect (2, –7) in the y-axis. → y-axis: the x-coordinate changes sign; (2, –7) → (–2, –7); quadrant: 4th

→ 3rd

Questions (in order: Easy, Medium, Challenging)

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|------------|-------------------|-------------------|
| 1. 2nd | 16. 4th | 31. (3, –4), 4th |
| 2. 1st | 17. 3rd | 32. (4, 3), 1st |
| 3. 4th | 18. 1st | 33. (–2, –3), 3rd |
| 4. 3rd | 19. 1st | 34. (–4, –5), 3rd |
| 5. 3rd | 20. 3rd | 35. (1, –3), 4th |
| 6. 1st | 21. 1st | 36. (4, –4), 4th |
| 7. 3rd | 22. 4th | 37. (–4, 2), 2nd |
| 8. 2nd | 23. 2nd | 38. (–2, 4) |
| 9. 3rd | 24. y-axis | 39. (–8, –4) |
| 10. x-axis | 25. 3rd | 40. (–5, –7) |
| 11. 4th | 26. 2nd | 41. (3, 6) |
| 12. 1st | 27. 4th | 42. (–6, 7) |
| 13. 1st | 28. 3rd | 43. (4, 8) |
| 14. 4th | 29. 1st | 44. (7, 4) |
| 15. 2nd | 30. (–1, –3), 3rd | |

9.04 Plotting tables of values (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: $x = 0$, $y = 5$ → (0, 5)
- Column 1, E2: $x = 6$, $y = 5$ → (6, 5)
- Column 2, E1: (1, 8), (2, 13), (3, 18) → y goes up by 5; next: (4, 23)
- Column 2, E2: (0, 2), (1, –1), (2, –4) → y goes down by 3; next: (3, –7)
- Column 3, E1: Plot (0, 1), (1, 4), (2, 7), (3, 10). → the points make a straight line; each time x goes up 1, y goes up 3
- Column 3, E2: Plot (0, 2), (1, 3), (2, 4), (3, 5). → the points make a straight line; each time x goes up 1, y goes up 1

Questions (in order: Easy, Medium, Challenging)

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|-------------|--------------|-------------------------------------|
| 1. (–1, 7) | 15. (5, 12) | 29. (0, 12) |
| 2. (–2, –4) | 16. (3, 12) | 30. (5, 3) |
| 3. (2, 0) | 17. (–3, 1) | 31. (4, 12) |
| 4. (–2, 0) | 18. (0, 9) | 32. (3, 11) |
| 5. (4, 6) | 19. (2, –3) | 33. (5, 16) |
| 6. (1, 12) | 20. (5, 9) | 34. (3, 18) |
| 7. (0, 11) | 21. (0, 8) | 35. (3, 4) |
| 8. (4, –4) | 22. (–3, 5) | 36. (5, 22) |
| 9. (–2, –2) | 23. (3, 3) | 37. (5, 17) |
| 10. (0, –4) | 24. (5, 1) | 38. (6, 31) |
| 11. (4, 2) | 25. (2, 4) | 39. (5, –14) |
| 12. (6, 9) | 26. (–1, 0) | 40. (3, 5) |
| 13. (3, 0) | 27. (5, 4) | 41. a straight line; y goes up by 1 |
| 14. (6, 10) | 28. (–2, –1) | |

9.05 Tables of values from rules (booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: $y = 5x + 6$, $x = 7$ → 41
- Column 1, E2: $y = 2x - 5$, $x = 1$ → –3
- Column 2, E1: $y = 4x + 8$ → $x = 0$: $y = 4 \times 0 + 8 = 8$; y: 8, 12, 16, 20
- Column 2, E2: $y = 5x$ → $x = 0$: $y = 5 \times 0 + 0 = 0$; y: 0, 5, 10, 15
- Column 3, E1: x: 0, 1, 2, 3
y: 0, 6, 12, 18 → y goes up by 6, so 6x; when $x = 0$, $y = 0$; $y = 6x$
- Column 3, E2: x: 0, 1, 2, 3
y: 5, 10, 15, 20 → y goes up by 5, so 5x; when $x = 0$, $y = 5$; $y = 5x + 5$

Questions (in order: Easy, Medium, Challenging)

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|-------|-------|------|
| 1. 33 | 2. 20 | 3. 2 |
|-------|-------|------|

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|--------|-------------------|-------------------|
| 4. 21 | 18. 0 | 32. 7, 9, 11, 13 |
| 5. 29 | 19. 11 | 33. 3, 5, 7, 9 |
| 6. 12 | 20. 31 | 34. 4, 9, 14, 19 |
| 7. 23 | 21. 15 | 35. 0, 1, 2, 3 |
| 8. 22 | 22. 38 | 36. 9, 14, 19, 24 |
| 9. -2 | 23. 20 | 37. 8, 11, 14, 17 |
| 10. 17 | 24. 13 | 38. $y = 6x + 6$ |
| 11. 11 | 25. 12 | 39. $y = 5x + 7$ |
| 12. 55 | 26. 46 | 40. $y = x + 8$ |
| 13. 26 | 27. 10 | 41. $y = x + 9$ |
| 14. 13 | 28. 3 | 42. $y = 4x + 7$ |
| 15. 23 | 29. 1, 4, 7, 10 | 43. $y = 5x$ |
| 16. 17 | 30. 7, 12, 17, 22 | |
| 17. -2 | 31. 6, 9, 12, 15 | |

9.06 Graphing linear equations (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: (3, 7); $y = 2x + 1 \rightarrow$ yes
- Column 1, E2: (-2, 1); $y = x + 3 \rightarrow$ yes
- Column 2, E1: $y = 2x - 2 \rightarrow y: -4, -2, 0$; points: (-1, -4), (0, -2), (1, 0)
- Column 2, E2: $y = -3x + 1 \rightarrow y: 4, 1, -2$; points: (-1, 4), (0, 1), (1, -2)

- Column 3, E1: Graph $y = 2x - 1 \rightarrow$ points: (-1, -3), (0, -1), (1, 1); cuts the y-axis at (0, -1)
- Column 3, E2: Graph $y = -2x - 2 \rightarrow$ points: (-1, 0), (0, -2), (1, -4); cuts the y-axis at (0, -2)

Questions (in order: Easy, Medium, Challenging)

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|---------|------------------------------|--------------------------------|
| 1. no | 18. yes | 32. (-1, 3), (0, 2), (1, 1) |
| 2. yes | 19. yes | 33. (-1, -3), (0, -2), (1, -1) |
| 3. yes | 20. no | 34. (-1, 1), (0, 4), (1, 7) |
| 4. yes | 21. yes | 35. (-1, -2), (0, -1), (1, 0) |
| 5. no | 22. no | 36. (-1, 3), (0, 4), (1, 5) |
| 6. no | 23. yes | 37. (-1, 4), (0, 2), (1, 0) |
| 7. yes | 24. yes | 38. (-1, 3), (0, 1), (1, -1) |
| 8. no | 25. yes | 39. (-1, 6), (0, 4), (1, 2) |
| 9. yes | 26. no | 40. cuts the y-axis at (0, 3) |
| 10. no | 27. yes | |
| 11. yes | 28. yes | |
| 12. yes | 29. yes | |
| 13. yes | 30. (-1, 0), (0, 1), (1, 2) | |
| 14. no | 31. (-1, -1), (0, 2), (1, 5) | |
| 15. yes | | |
| 16. yes | | |
| 17. no | | |